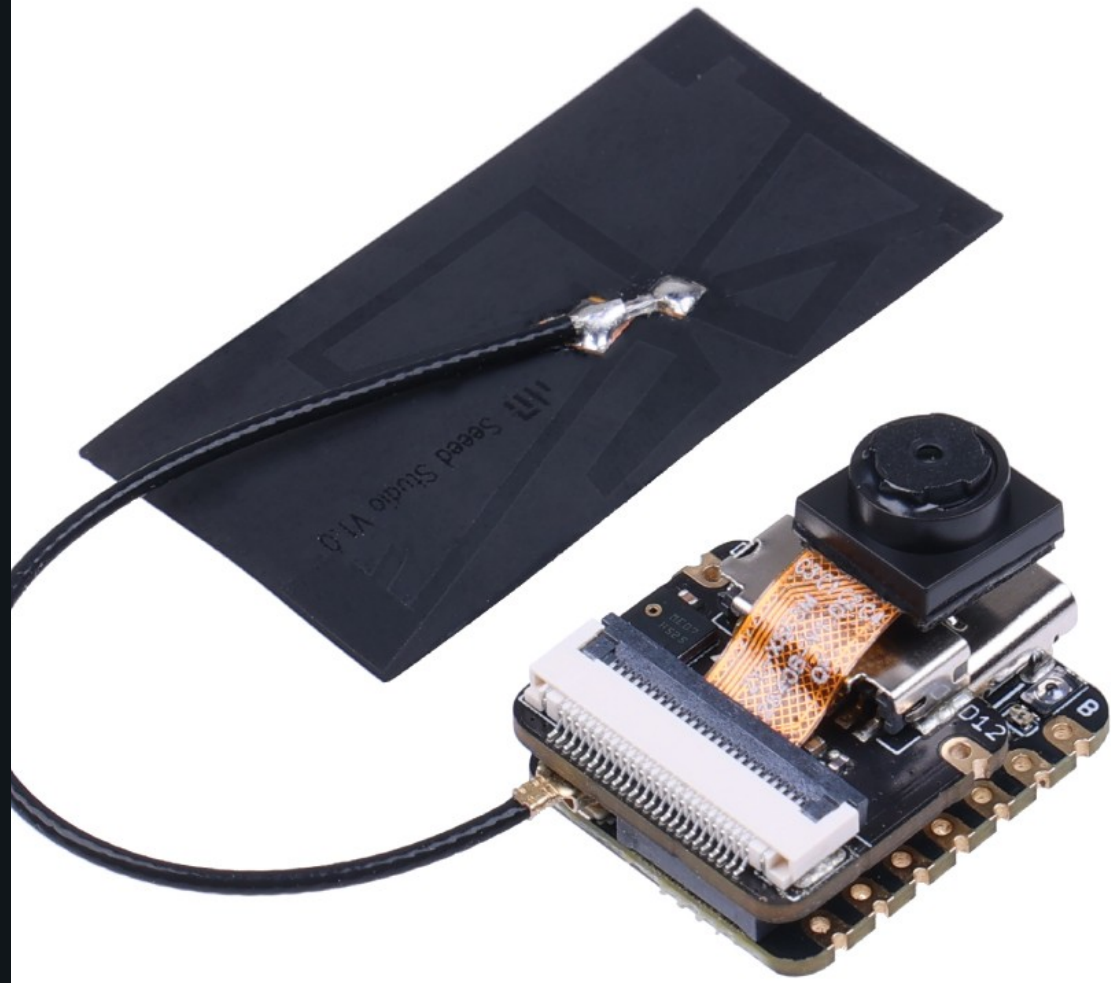
ECE 116

Advanced Hardware & Software Design

Intelligent embedded systems | Fall 2026



Instructional Team

Course staff, office hours, room, and meeting time are published on Canvas.



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Intelligent systems live where software meets the physical world



health



exploration



assistive care



automation

Different applications. The same need to connect computation to real conditions and consequences.

A useful device must observe, decide, and act

observe a request or condition

maintain state and make a decision



act in the physical world

make the result observable

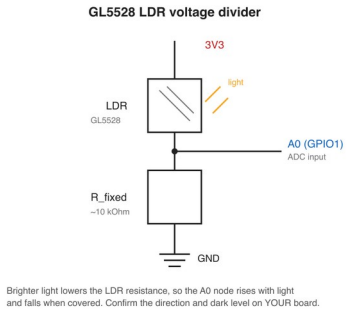
The difficult bugs often live in the interfaces between these responsibilities.

XIAO ESP32-S3 Sense



- camera
- microphone
- GPIO, ADC, and buses
- USB power, upload, and serial

Assignments and final project



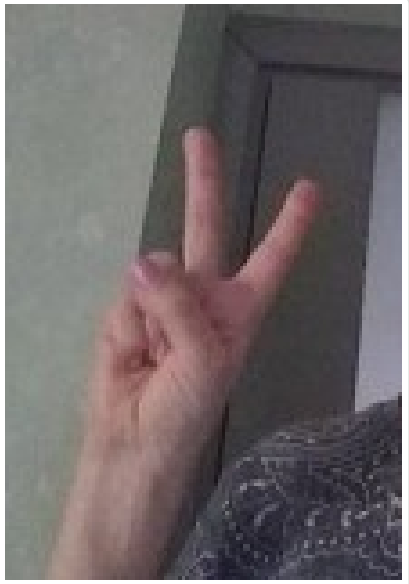
A1

Microcontroller & Finite State Machine



A2

Evaluating a Provided Voice Model



A3

Training a Vision Model



A4

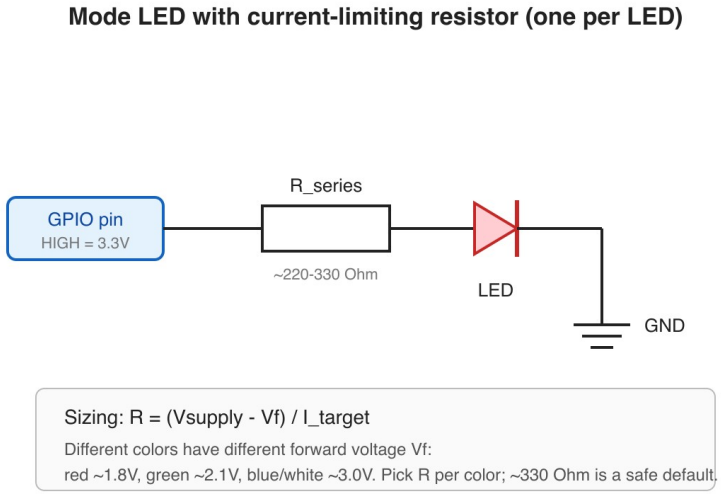
Vision Deployment Under Constraints



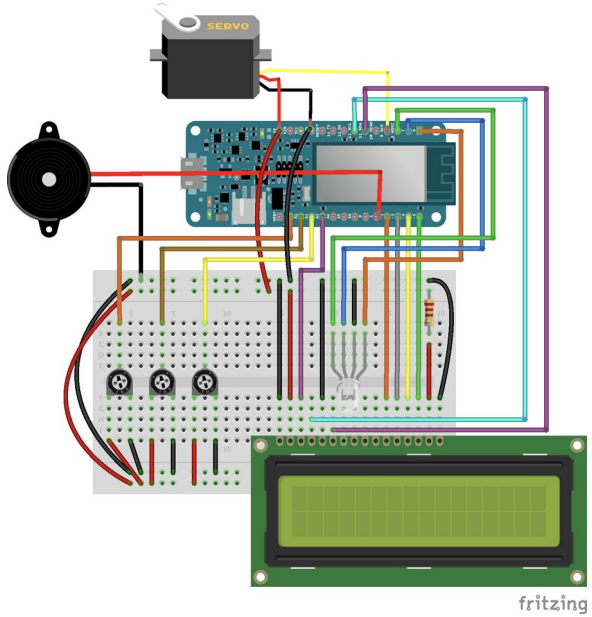
Project

Local ML gate and useful action

A1–A4: labs and challenges



In-class labs: controlled mechanism



Challenge: new system context

Illustrations show the lab/challenge pattern. Use each handout for the required hardware.

Class Schedule

Both days: 9:30 a.m.–12:20 p.m. Pacific

Tuesday

LECTURE
(Zoom)

Conceptual overviews

Thursday

LAB OR OPEN WORK
Jacobs Hall, B538

Bring the XIAO, laptop, a known-good data cable, and the components required for hardware work.

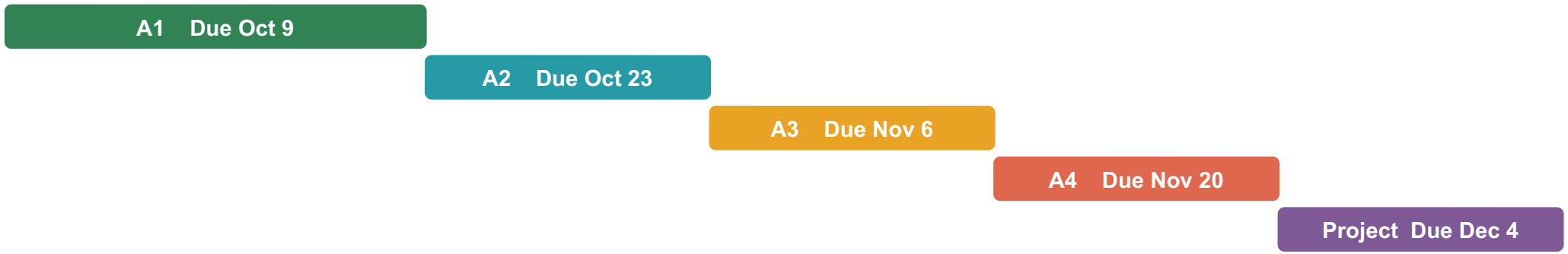
First meeting: Thu Sep 24, in person. L00 + L01 and Lab 0.

Course Cadence

Lectures introduce the next topic during the current assignment's due week.

	W0 Sep 22	W1 Sep 29	W2 Oct 6	W3 Oct 13	W4 Oct 20	W5 Oct 27	W6 Nov 3	W7 Nov 10	W8 Nov 17	W9 Nov 24	W10 Dec 1
LECTURE Tuesday / Zoom	No class	L02 L03	L04 L05	L06 L07	L08 L09	L10 L11	L12 L13	L14 L15	L16 Project intro	Project work	Project work
LAB / WORK Thursday In person	Lecture L00 + L01 + Lab 0 Sep 24	A1 lab Oct 1	A1 work Oct 8	A2 lab Oct 15	A2 work Oct 22	A3 lab Oct 29	A3 work Nov 5	A4 lab Nov 12	A4 work Nov 19	No class Nov 26	Project work Dec 3

ASSIGNMENT BUILD WINDOWS



Deadlines: Friday 5 p.m. Pacific. Next assignment releases Monday.

Course Grade Components

Assignments

75%

A1	A2	A3	A4
18.75%	18.75%	18.75%	18.75%

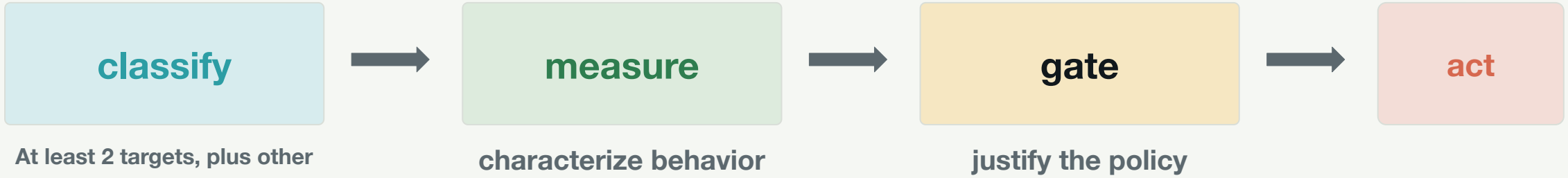
Final project

25%

Final project
25%

Report: 4 technical + 1 AI-use + 1 reference page
Video demo + trial-data ZIP; submit all three files

Final project



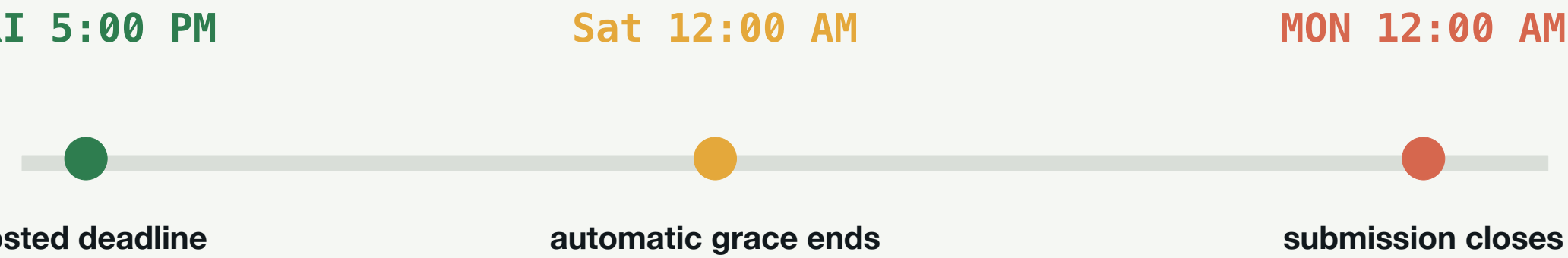
Local sensing and ML gate a useful action. Due Dec 4.
Report (4+1+1 pages), video, and trial-data ZIP through Canvas.

Finding course materials in Canvas

I need to…	Open…
Get set up for the course	**Home → Start Here**
Find this week's slides and preparation	**Modules → Current week**
Check the quarter schedule or policies	**Syllabus**
Download an assignment, read its rubric, or submit	**Assignments → A1–A4 + Project**
See course updates	**Announcements**
Find grades and feedback	**Grades**

A1–A4: one ZIP. Final project: report PDF + MP4 + trial-data ZIP.
Handout and generic Word template: Start Here, available from Week 0.

Late Policy



No late deduction
Through Fri 11:59 p.m.

Deduct 25% of earned score
Saturday and Sunday

Submission closed
From Mon 12 a.m.

All times Pacific. Approved individual accommodations and extensions apply.

Collaboration Policy

WORK TOGETHER

discuss concepts
compare test strategies
interpret error messages
work near classmates

TAKE RESPONSIBILITY

design decisions
your own measurements
verified evidence
clear attribution

Do not exchange ready-to-submit work or use another student's measurements as your own. University academic integrity policies apply.

AI Use in This Course

Allowed

- Ask for an explanation of a concept
- Debug a specific error or compare approaches
- Final project: generate and adapt code
- Test, revise, and explain assisted work

Not allowed

- Submit generated work you cannot explain
- Fabricate measurements, screenshots, or evidence
- Present untested output as verified
- Bypass collaboration or citation requirements

Disclose assistance in your README or project report: tool, contribution, and verification.

Lab 0: Kit setup and bring-up

Collect your kit. Solder headers with staff and inspect before powering the board.



compile

upload

serial

LED

Bring your laptop and a USB data cable. Lab 0 has no graded submission.